

Assignment #3

Files4A3.zip contains three java files, one .jpg file, one Blender file ([link](#) for original file), and two .obj files.

1. Within your package `codes**8490`, create three new java files whose names are the same as the provided java files, except that **XY** in the names need to be replaced with your initials.
 - (a) Copy codes from the provided files to your corresponding files;
 - (b) Extract and place the .jpg and .obj files to the folders accessible from the codes;
 - (c) Run your `A3_**.java` file should produce a result as shown in the lower left image.
2. Open the Blender file to extract and save the additional four components (i.e., **Shaft**, **Motor**, **Blade**, and **Guard**) as separate files to the same folder of the provided .obj files.
3. Modify your `A30bjects**.java` file to create a derived class for each of the four additional components, and code your `A3_**.java` file to render a result as illustrated in the (middle and right) figures below.
 - (a) Complete the structure by placing the additional components at their right locations and orientations and linking them one by one with **TransformGroups**.
Hint: The suggested scaling values are 0.18, 3.5, 2.9, and 3.3 for **Shaft**, **Motor**, **Blade**, and **Guard** respectively.
 - (b) Set all components with the same appearance as the **Stand** but use different colors for the **Shaft** and the connecting cylinder of the **Blade** as shown in the figures.
 - (c) While the base remains stationary, the blades (and the connecting cylinder) rotate counterclockwise continuously at a speed of 500 milliseconds per round rotation.
 - (d) The cylindrical **Shaft** rotates in such way that it make the **Motor** (and all the other components attached to it) oscillate back and forth from one side to another continuously.
 - (e) It takes 5,000 milliseconds for the **Motor** to reach from one side to the other, with the ramp duration set to 2,500 milliseconds. After reaching to one end, the **Motor** stays for 200 milliseconds before starting to rotate back.
Hint: the specification of **Alpha** class is available at this [link](#).
 - (f) Use the **z** key as a pause key to pause or resume the **Motor**'s oscillation, and use the **x** key as the power key to stop or resume both the **Motor**'s oscillation and **Blade**'s rotation. When the desktop fan is powered on again (with a press of the **x** key), the **Motor**'s oscillation and **Blade**'s rotation resume from the location and in the direction when the fan was powered off.
 - (g) Update the string label on the base to identify your work, and make your program easy to comprehend by adding adequate amount of comments.
4. Locate the folder that contains all files of your project `Comp8490`; produce a zip file of your project folder; and submit it online before due.

